



MSc Technologies and Analytics in Precision Medicine

The Identification of Novel PPAR γ Modulators from a Marine-Derived Chemical Space

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Abstract

Inflammatory bowel disease (IBD) is a chronic inflammatory disorder of the gastrointestinal tract comprising Crohn's disease and ulcerative colitis, and patients with long-standing IBD have a significantly increased risk of developing colorectal cancer (CRC). Peroxisome proliferator-activated receptor gamma (PPAR γ) is highly expressed within colonic epithelial cells and represents an attractive therapeutic target, but long-term clinical application of synthetic agonists is limited by adverse effects including weight gain, oedema, cardiovascular complications and increased fracture risk. The aim of this project was to identify and prioritise novel seaweed-derived PPAR γ modulators from *Saccharina latissima*, *Laminaria digitata* and *Undaria pinnatifida* using rational computational drug discovery and AI-assisted screening approaches for the treatment of IBD and CRC.

A reference database of experimentally validated PPAR γ agonists was curated from the Protein Data Bank and integrated with surface mapping, ligand interaction analysis and structure–activity relationship analysis to generate a consensus pharmacophore, which was then used alongside Tanimoto similarity screening, AI-assisted prioritisation and molecular docking to screen a curated database of 51 brown seaweed metabolites.

Undaria pinnatifida exhibited the greatest structural similarity to known PPAR γ agonists and contained the majority of pharmacophore hits, and fatty acids consistently emerged as the highest-priority candidates across all three independent screening methods, with all three consensus compounds ranking within the ten highest-scoring docked ligands.

These findings support the hypothesis that marine-derived metabolites possess structural characteristics compatible with PPAR γ ligand recognition and demonstrate the value of integrating complementary computational approaches for ligand prioritisation, providing a framework for the future discovery of marine-derived PPAR γ modulators for inflammatory disease research

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Abbreviations

AF-2	Activation Function-2
ACD	Arachidonic acid
AI	Artificial Intelligence
CADD	Computer-Aided Drug Design
CD	Crohn's Disease
ChEMBL	Chemical Entities of Biological Interest
CRC	Colorectal Cancer
Da	Dalton
DNA	Deoxyribonucleic Acid
EC ₅₀	Half Maximal Effective Concentration
EPA	Eicosapentaenoic Acid
FXR	Farnesoid X Receptor
HBA	Hydrogen Bond Acceptor
HBD	Hydrogen Bond Donor
IBD	Inflammatory Bowel Disease
LBD	Ligand-Binding Domain
LogP	Octanol–Water Partition Coefficient
LOTUS	natural prOducTs occUrrences databaSe
MOE	Molecular Operating Environment
NF- κ B	Nuclear Factor kappa B
PDB	Protein Data Bank
PPAR γ	Peroxisome Proliferator-Activated Receptor Gamma
RCSB	Research Collaboratory for Structural Bioinformatics
RDKit	RDKit Cheminformatics Toolkit
RMSD	Root Mean Square Deviation
RXR	Retinoid X Receptor
SAR	Structure–Activity Relationship
SMARTS	SMILES Arbitrary Target Specification
SMILES	Simplified Molecular Input Line Entry System
TPSA	Topological Polar Surface Area
TZD	Thiazolidinedione
UC	Ulcerative Colitis
UMAP	Uniform Manifold Approximation and Projection

UniProt Universal Protein Resource

1 Introduction

1.1 Inflammatory Bowel Disease and Colorectal Cancer

Inflammatory bowel disease (IBD) is a chronic inflammatory disorder of the gastrointestinal tract comprising Crohn's disease (CD) and ulcerative colitis (UC), where CD is able to affect any region of the gastrointestinal tract and UC primarily restricted to the colon and rectum [1]. IBD pathogenesis is multifactorial, involving genetic susceptibility, immune dysregulation, environmental factors, and persistent inflammatory signalling which drives epithelial damage, oxidative stress and cytokine production that increase susceptibility to colorectal cancer (CRC) development [2]. IBD incidence and prevalence continue to rise globally, placing a substantial burden on healthcare systems [3], and approximately 40,000 individuals are currently affected by IBD in Ireland [4]. Patients with long-standing IBD have a significantly increased risk of developing CRC, the third most common cancer globally, responsible for approximately 12.4% of all cancer-related deaths [5]. As IBD requires lifelong disease management, identifying safe long-term therapeutic strategies remains an important clinical priority.

1.2 FXR Signalling and Therapeutic Limitations

The farnesoid X receptor (FXR) is a bile acid-activated nuclear receptor that maintains intestinal homeostasis through regulation of epithelial barrier integrity, bile acid metabolism and inflammatory signalling [6]. Reduced intestinal FXR expression has been reported in both IBD and CRC, with receptor deficiency associated with increased inflammation and tumour formation [7]. However, full FXR activation is associated with adverse metabolic effects relating to bile acid and cholesterol regulation, prompting interest in alternative strategies capable of retaining beneficial FXR activity while reducing mechanism-associated side effects [8].

1.3 PPAR γ as an Alternative Therapeutic Target

Peroxisome proliferator-activated receptor gamma (PPAR γ) is one of three PPAR family members, alongside PPAR α and PPAR δ , and is highly expressed within adipose tissue, macrophages and colonic epithelial cells, where it regulates lipid metabolism, glucose homeostasis and inflammatory signalling [9][10]. The high expression of PPAR γ within colonic tissue makes it an attractive therapeutic target for ulcerative colitis, where inflammation is primarily localised to the colon [11]. PPAR γ contains a conserved DNA-binding domain and a C-terminal ligand-binding domain (LBD) that heterodimerises with the retinoid X receptor (RXR) following ligand binding, stabilising the activation function-2 (AF-2) region and promoting recruitment of transcriptional co-activator proteins [12][10]. PPAR γ also suppresses inflammatory signalling through transrepression of NF- κ B-mediated pathways [13] and increases FXR

gene expression [14], suggesting that PPAR γ modulation may indirectly restore intestinal signalling disrupted in IBD and CRC.

Synthetic PPAR γ agonists, particularly thiazolidinediones (TZDs), exhibit anti-inflammatory and insulin-sensitising properties and have indicated therapeutic potential in inflammatory disease [celinski_2_2012]. However, long-term clinical application is limited by adverse effects including weight gain, oedema, cardiovascular complications and increased fracture risk [12], driving considerable research interest towards selective or partial PPAR γ agonists capable of retaining therapeutic efficacy while improving safety profiles [14]. To date, this research has focused almost exclusively on synthetic chemotypes, leaving natural product-derived scaffolds – despite their generally favourable safety profiles – comparatively unexplored as a source of selective PPAR γ modulators.

1.4 Marine-Derived Natural Products and Computational Drug Discovery

Natural products represent an important source of structurally diverse bioactive compounds with favourable drug-like physicochemical properties [15], and marine-derived compounds in particular possess significant anti-inflammatory and immunomodulatory activity, making them promising candidates for intestinal disease therapeutics [16]. Brown seaweeds including *Saccharina latissima*, *Laminaria digitata* and *Undaria pinnatifida* contain structurally diverse metabolites with favourable biocompatibility, biodegradability and low toxicity [17], and several seaweed-derived compounds reduce inflammatory cytokine production, improve epithelial barrier integrity and ameliorate symptoms in experimental models of colitis [16] – making these species a biologically plausible, but unexplored, source of safer PPAR γ modulators.

Traditional natural product drug discovery is expensive and time-consuming, so computer-aided drug design (CADD) – including pharmacophore modelling, molecular similarity analysis and structure-based docking – is increasingly used to prioritise candidates before experimental validation [18], with AI-assisted screening further enhancing virtual discovery by predicting bioactivity directly from chemical structure [19].

1.5 Aims and Objectives

The overall aim of this study was to identify and prioritise novel seaweed-derived PPAR γ modulators from *Saccharina latissima*, *Laminaria digitata* and *Undaria pinnatifida* using rational computational drug discovery and AI-assisted screening approaches for the treatment of inflammatory bowel disease (IBD) and colorectal cancer (CRC). The specific objectives were to:

1. Characterise the structural diversity of known PPAR γ agonists.
2. Characterise the PPAR γ ligand-binding pocket.

3. Determine the structural features associated with PPAR γ agonism.
4. Apply computational screening approaches to the seaweed metabolite database.
5. Validate prioritised compounds using molecular docking.

2 Materials and Methods

2.1 PPAR γ agonist and seaweed metabolite database curation

A reference database of experimentally validated PPAR γ agonists was curated from the RCSB Protein Data Bank [20] using UniProt [21] accession P37231, selecting *Homo sapiens* X-ray crystal structures with a resolution ≤ 2.5 Å. Structures were filtered to remove crystallographic artefacts and ligands below 150 Da, annotated with EC₅₀ values from ChEMBL [22], BindingDB [23] and primary literature, classified as wild-type or mutant, and deduplicated by ligand identifier, SMILES representation and mutation status. A brown seaweed metabolite database was compiled from the LOTUS Natural Products database [24] for *Saccharina latissima*, *Laminaria digitata* and *Undaria pinnatifida*, excluding compounds lacking a SMILES structure or below 150 Da. Molecular descriptors (molecular weight, LogP, TPSA, hydrogen bond donors/acceptors, rotatable bonds, aromatic rings and heavy atoms) were calculated in RDKit for both databases.

2.2 Chemotype discovery and structural classification

Morgan fingerprints (radius 2, 2048-bit, chirality enabled) were generated from SMILES structures and clustered using the Butina algorithm with distance cut-offs of 0.60 (agonists) and 0.40 (seaweed). Cluster quality was assessed using two-dimensional UMAP visualisation ($n = 15$ nearest neighbours, minimum distance = 0.1), silhouette score and intra-cluster similarity. Representative compounds from each cluster were compared with literature-reported chemotypes, encoded as SMARTS patterns and applied as substructure queries to both databases, with unmatched compounds classified as Other/Unclassified.

2.3 Binding pocket characterisation

Representative PPAR γ -ligand complexes (93 total; 51 with EC₅₀ data) spanning all chemotypes were analysed in Molecular Operating Environment (v2020.09) [25] using the Compute Ligand Interactions module to identify hydrogen-bond donor, hydrogen-bond acceptor, ionic and π -H interactions. Interaction data were consolidated in Python (Pandas), and residue frequencies calculated for each chemotype. Residues present in at least three chemotypes were defined as consensus residues; Tyr473 was additionally retained because of its established role in PPAR γ activation. Molecular surface representations of binding pocket accessibility, lipophilicity and electrostatic potential were generated in MOE to characterise the physicochemical environment and support pharmacophore construction.

2.4 Structure–activity relationship analysis

Experimentally determined EC_{50} values were converted to pEC_{50} , and ligands classified as high ($pEC_{50} \geq 6.5$) or low ($pEC_{50} < 6.5$) activity. Representative ligands were visualised in MOE, and functional groups (carboxylic acids, hydroxyls, amides, sulfonamides, ethers, ketones, aromatic/heteroaromatic rings, halogens, trifluoromethyl groups, flexible linkers and double bonds) manually annotated and compared between activity groups to identify structural features associated with agonist potency.

2.5 Pharmacophore generation

Pharmacophore features were defined by integrating the surface mapping, ligand interaction and SAR analyses. Features were manually assigned to the co-crystallised ligand of a representative PPAR γ structure (PDB ID: 5Z6S) using the MOE Pharmacophore Query Editor and verified using the Pharmacophore Elucidation module. The final pharmacophore comprised hydrogen-bond acceptor, hydrogen-bond acceptor/negative ionisable, aromatic and hydrophobic features representing the conserved PPAR γ interaction network. Volume exclusion spheres were incorporated to define sterically inaccessible regions of the binding pocket, and the completed model exported as a .ph4 query for virtual screening.

2.6 Computational screening strategies for seaweed metabolite prioritisation

Tanimoto similarity coefficients (Morgan fingerprints, radius 2, 2048-bit) were calculated between representative reference agonists and all seaweed metabolites and aggregated by seaweed species and agonist class (Non-TZD synthetic, TZD synthetic, Natural Product Derived and Fatty Acid/Lipid-Like). For pharmacophore screening, seaweed metabolites were washed, tautomer-enumerated and conformationally expanded in MOE using LowModeMD (≤ 250 conformers per compound) before screening against the consensus pharmacophore using the Best per Molecule option. A minimum match threshold of five of nine pharmacophore features was used for virtual screening, with RMSD recorded as a measure of pharmacophore fit. For comparison with molecular docking, a second search using a minimum threshold of three matched features was performed to generate RMSD values for all seaweed metabolites.

AI-assisted prioritisation was performed using ChatGPT [26]. Each metabolite was evaluated from its SMILES structure, molecular weight, LogP, TPSA, hydrogen-bond donor/acceptor counts and rotatable bond count, and assigned a High, Medium or Low priority with an associated confidence score and structural rationale. Results from Tanimoto similarity, pharmacophore screening and AI prioritisation were integrated using multivariate analysis in MOE to identify consensus candidates.

2.7 Molecular docking as an independent validation method

Molecular docking was performed in MOE using the Dock module. The docking site was defined from the co-crystallised ligand in the PPAR γ crystal structure (PDB ID: 5Z6S), comprising all residues within 4.5 Å of the ligand (Ile249, Leu255, Gly258, Glu259, Ile262, Arg280, Ile281, Gly284, Cys285, Arg288, Ser289, Ile326, Tyr327, Leu330, Val339, Leu340, Ile341, Met348, Leu353, Phe363, Met364, Lys367 and His449). Ligands were docked using the London dG scoring function with a wall constraint.

The protocol was validated by re-docking the co-crystallised ligand, generating five poses that were evaluated by docking score, recovery of the crystallographic binding mode and conservation of protein–ligand interactions relative to the experimental structure. Following validation, 51 seaweed metabolites were docked using identical receptor preparation and docking parameters. Docking performance was subsequently compared with Tanimoto similarity, pharmacophore screening and AI-assisted prioritisation to assess agreement between screening methods and identify additional high-ranking candidates.

2.8 Computational environment and data analysis

All cheminformatics, statistical analyses and data visualisation were performed in Python (v3.14) [27]. Cheminformatics analyses were conducted using RDKit (v2025.03.4) [28], data manipulation using Pandas [29], numerical analyses using NumPy [30] and statistical analyses using SciPy [31]. Figures were generated using Matplotlib [32] and Seaborn [33]. Pearson correlation coefficients were calculated to assess the relationship between pharmacophore root mean square deviation (RMSD) and molecular docking scores.

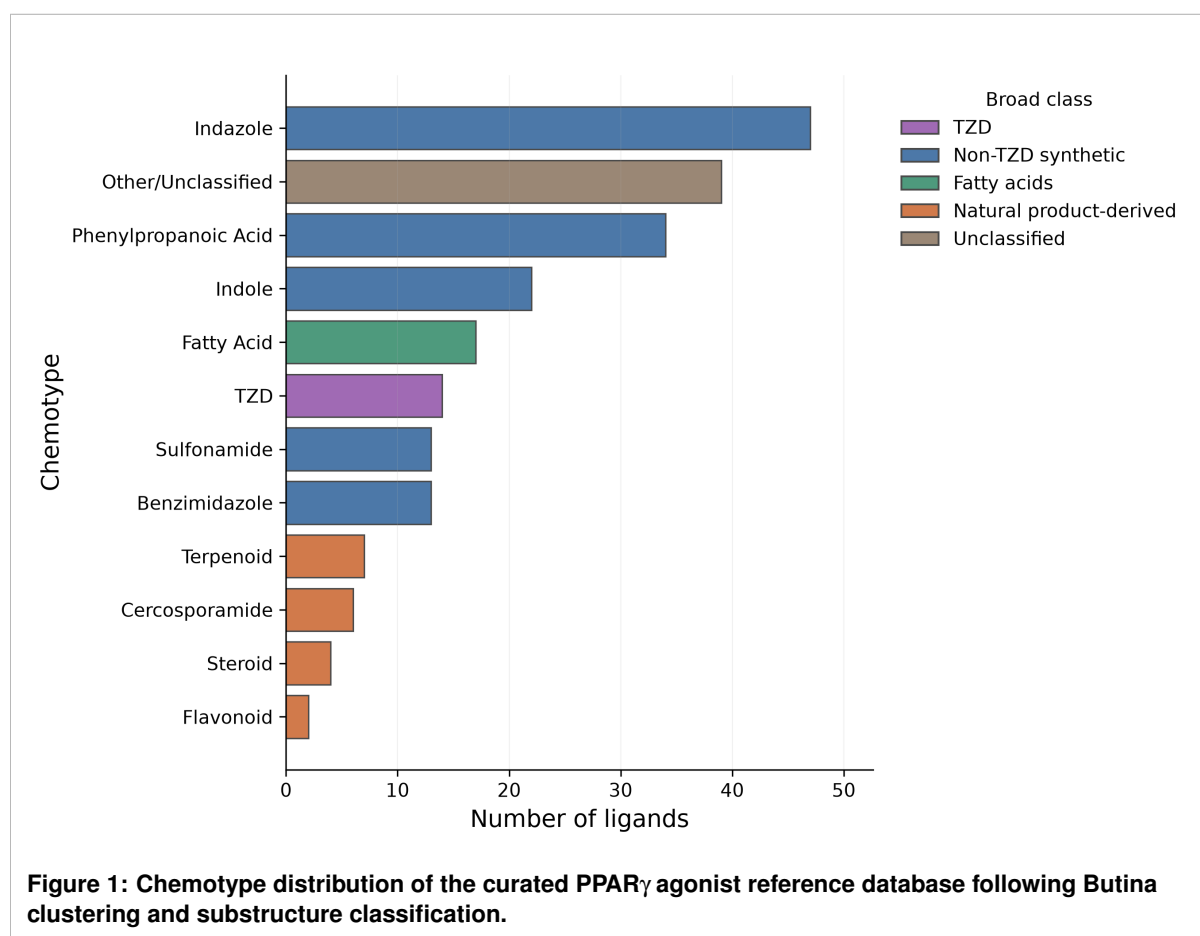
3 Results

3.1 Diverse chemotypes were identified within the curated PPAR γ agonist and seaweed metabolite databases

Chemotype discovery and structural classification were performed to characterise the chemical diversity of the curated PPAR γ agonist reference database and brown seaweed metabolite database prior to computational screening.

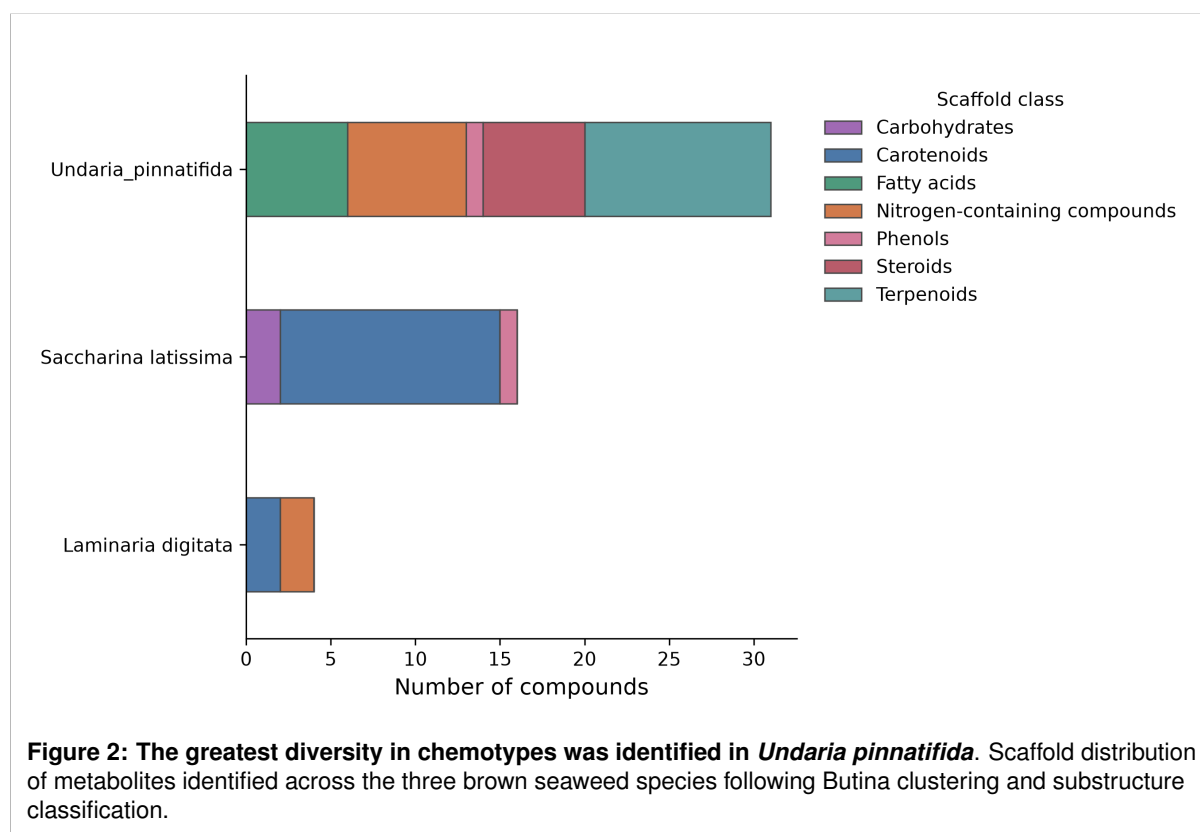
3.1.1 Chemotype Classification Reveals the Structural Diversity of PPAR γ Agonists

A total of 226 co-crystallised PPAR γ agonists were curated from the Protein Data Bank, comprising 218 wild-type and 8 mutant complexes, of which 66 had experimentally reported EC₅₀ values. Non-TZD synthetic ligands represented the predominant chemotype, with fatty acids, TZDs and natural product-derived agonists also identified, providing a chemically diverse reference dataset for subsequent analyses (Figure 1).



3.1.2 Chemotype Classification Reveals the Structural Diversity of Seaweed Metabolites

The curated seaweed metabolite database comprised 51 metabolites: 31 from *Undaria pinnatifida*, 16 from *Saccharina latissima* and 4 from *Laminaria digitata*. *Undaria pinnatifida* exhibited the greatest scaffold diversity across six classes, including fatty acids, nitrogen-containing compounds, steroids and terpenoids, whereas *Saccharina latissima* was largely dominated by carotenoids and *Laminaria digitata* contained representatives from only two scaffold classes (Figure 2).

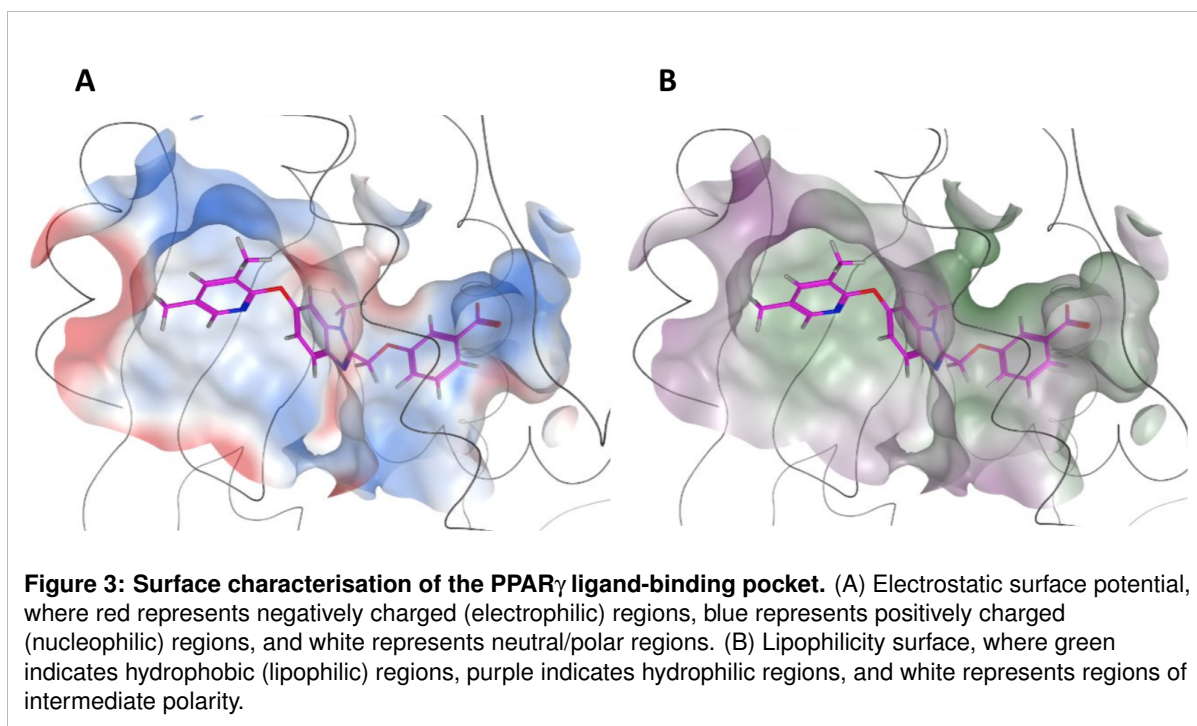


3.2 PPAR γ Binding Pocket Characterisation Reveals Conserved Structural Features for Ligand Recognition

Having established the structural diversity of both databases, the PPAR γ ligand-binding pocket itself was characterised to determine which physicochemical and residue-level features underlie receptor recognition.

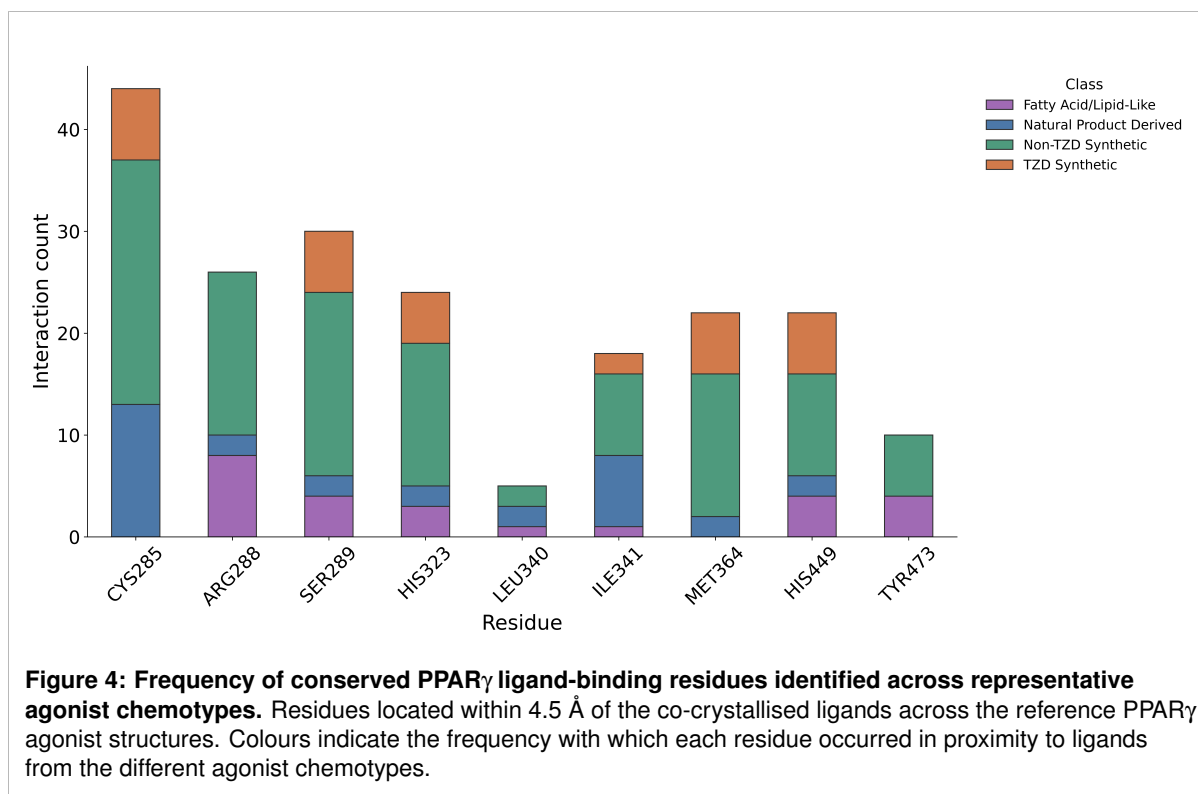
3.2.1 Conserved polar and hydrophobic interactions define the PPAR γ ligand-binding pocket

The PPAR γ ligand-binding cavity exhibited distinct negatively charged, positively charged and neutral electrostatic regions, together with a predominantly hydrophobic lipophilicity surface containing only limited hydrophilic regions (Figure 3), indicating that favourable PPAR γ agonists require a predominantly hydrophobic scaffold capable of forming complementary electrostatic interactions.



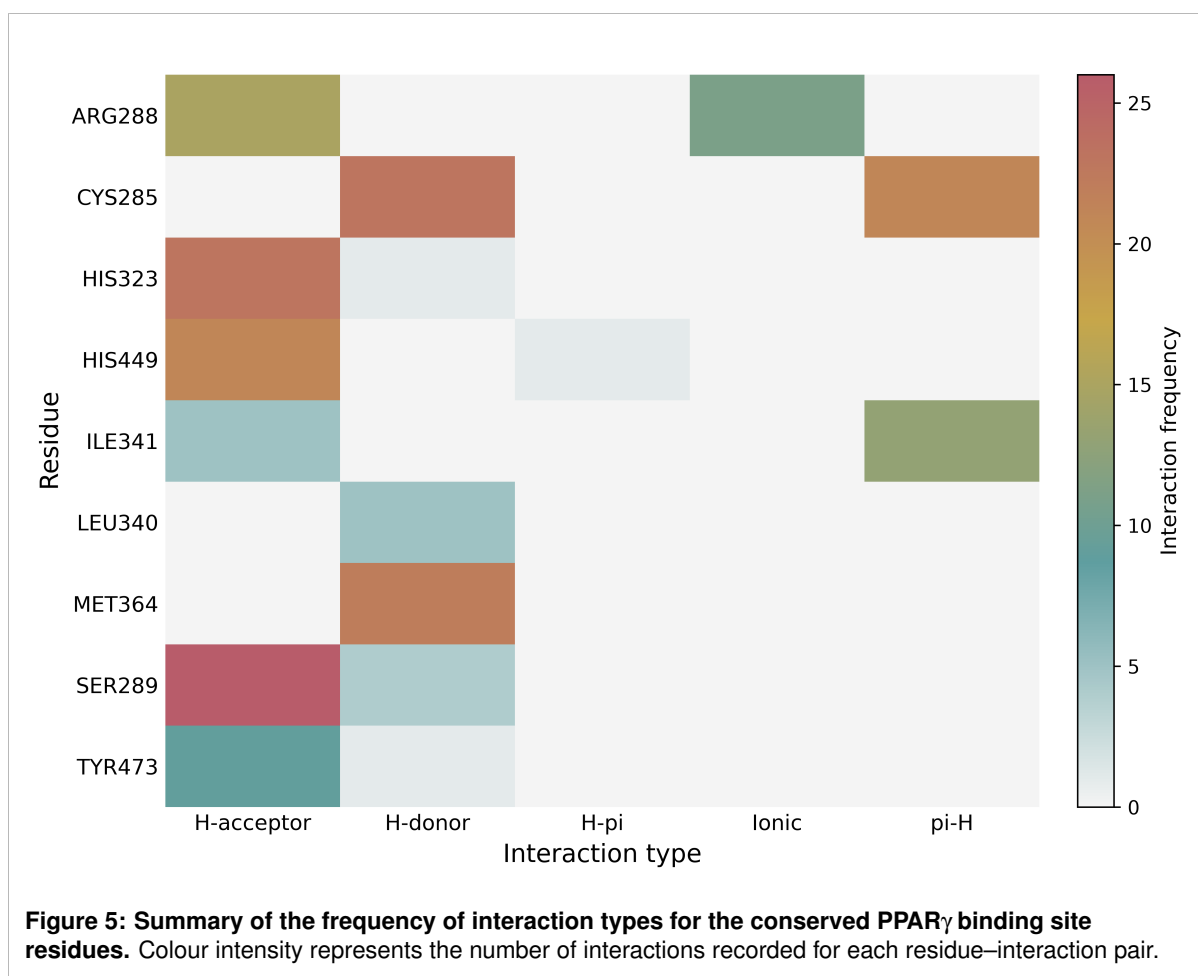
3.2.2 A conserved network of ligand-binding residues is shared across PPAR γ agonist chemotypes

CYS285 was the residue most frequently observed in close proximity to ligands across the major chemotypes, followed by SER289, ARG288, HIS323, MET364 and HIS449. LEU340, ILE341 and TYR473 were observed less frequently and were associated with a smaller subset of agonist classes (Figure 4).



3.2.3 Hydrogen-bonding interactions dominate PPAR γ ligand recognition

Hydrogen-bond acceptor interactions were most frequent for SER289, HIS323 and HIS449, whereas CYS285 and MET364 more frequently participated in hydrogen-bond donor interactions, while π -H interactions occurred less frequently and were largely confined to a subset of residues (Figure 5).

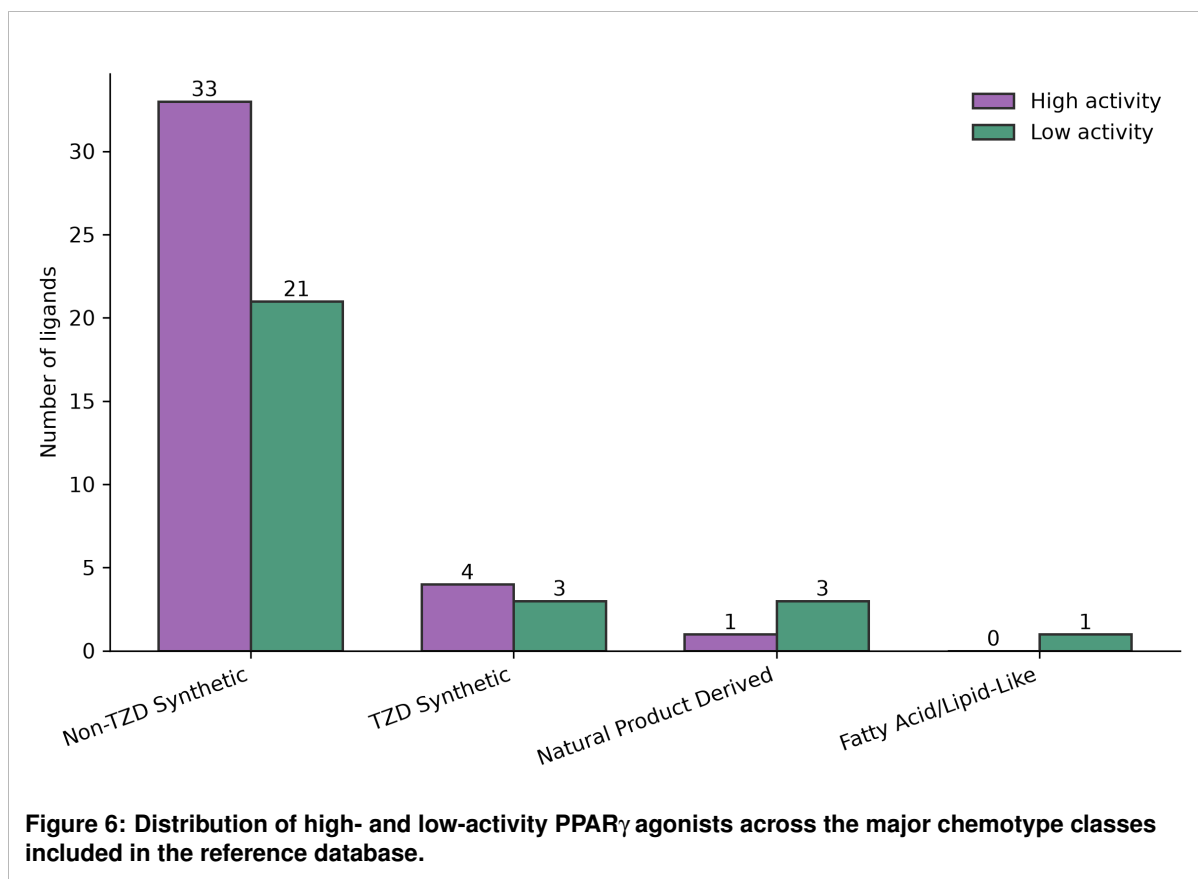


3.3 Integrated Structural Analyses Identify the Conserved Features of PPAR γ Agonism

These binding pocket features were next integrated with structure–activity relationship analysis, allowing the molecular determinants of PPAR γ agonism to inform construction of a consensus pharmacophore.

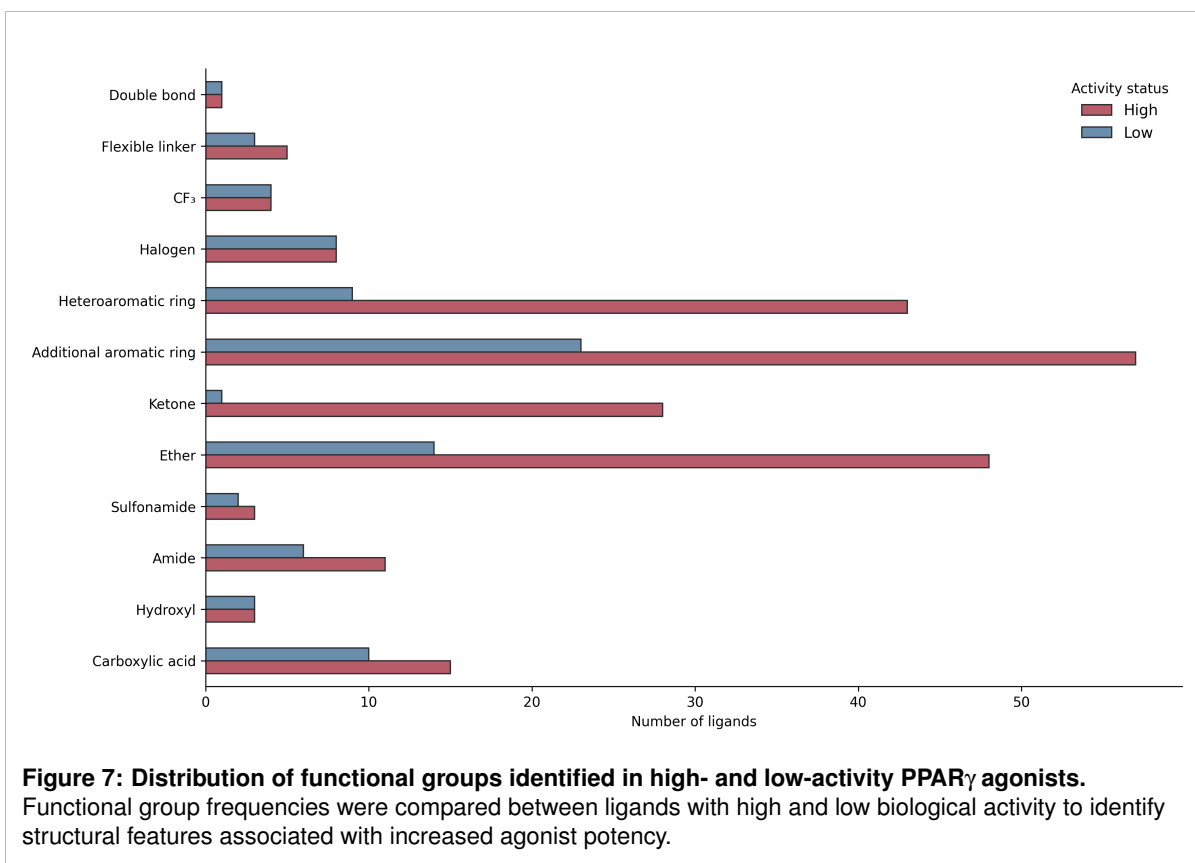
3.3.1 Highly active agonists are predominantly non-TZD synthetic compounds

Highly active agonists were predominantly non-TZD synthetic compounds, reflecting the overall composition of the reference database (Figure 6).



3.3.2 Highly active agonists share conserved aromatic and hydrogen-bonding features

These compounds exhibited a greater frequency of aromatic and heteroaromatic ring systems together with hydrogen-bonding functional groups, including ketones, ethers and carboxylic acids, than low-activity ligands (Figure 7).



3.3.3 Multiple complementary analyses support consensus pharmacophore feature selection

Integration of the surface mapping, ligand interaction and SAR findings generated a consensus nine-feature pharmacophore comprising four aromatic, four hydrogen-bond acceptor, three hydrophobic and two negative ionisable features, together with volume exclusion spheres defining sterically inaccessible regions of the binding pocket. (Figure 8).

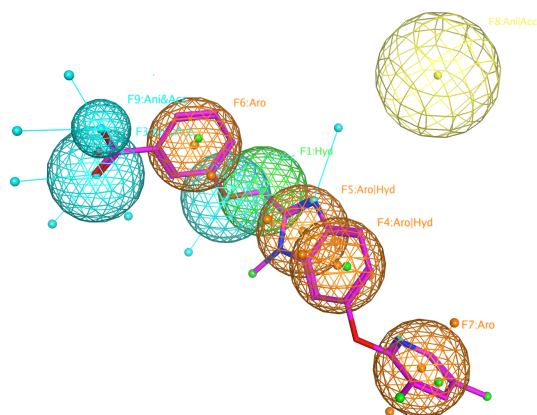


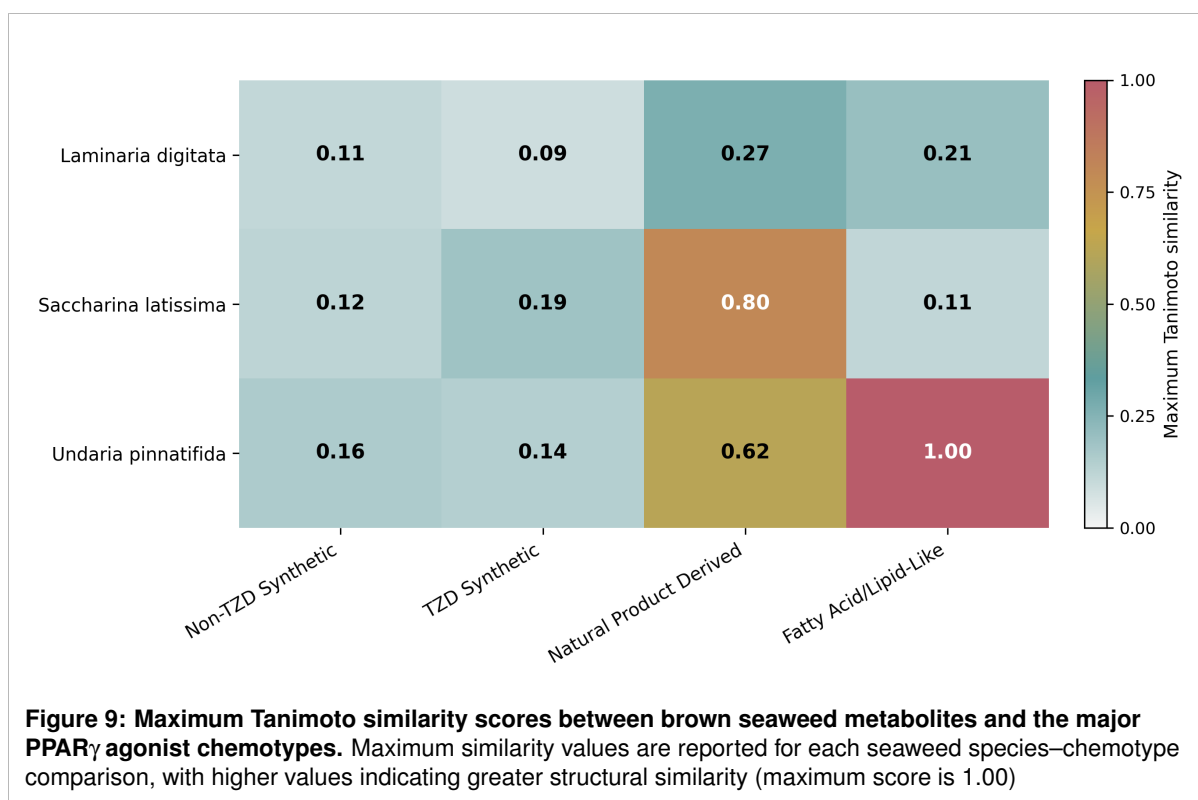
Figure 8: Consensus pharmacophore model representing the conserved structural requirements for PPAR γ agonism. Comprise of four aromatic features (Aro), four hydrogen bond acceptor features (Acc), three hydrophobic features (Hyd) and two negative ionisable features (Ani). The pharmacophore was generated using the 5S6Z PDB structure and the co-crystallised 5S6Z ligand is overlaid onto the pharmacophore

3.4 Complementary Computational Screening Approaches Prioritise Candidate PPAR γ Agonists from Brown Seaweed Metabolites

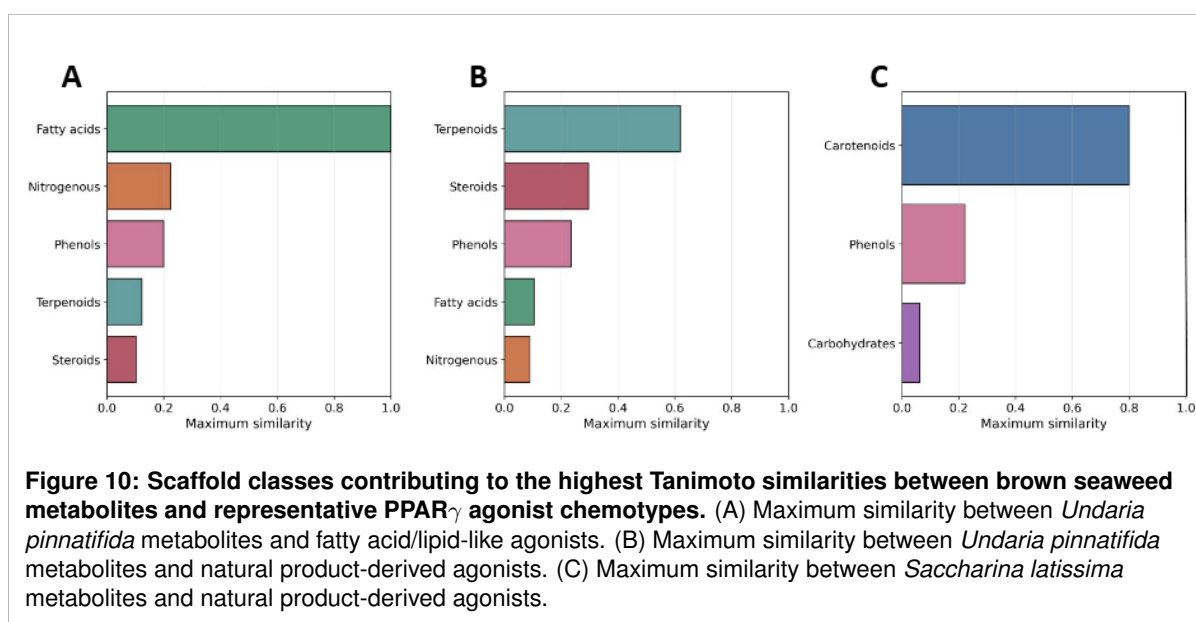
With a consensus pharmacophore established, three independent screening strategies – Tanimoto similarity, pharmacophore matching and AI-assisted prioritisation – were applied to the curated seaweed database to identify and rank candidate PPAR γ agonists.

3.4.1 *Undaria pinnatifida* exhibited the greatest structural similarity to known PPAR γ agonists

Undaria pinnatifida exhibited the strongest similarity to the reference agonists overall, including an exact structural match (Tanimoto similarity = 1.00) to a fatty acid/lipid-like agonist and a maximum similarity of 0.62 to natural product-derived agonists. *Saccharina latissima* showed substantial similarity to natural product-derived agonists (0.80); and *Laminaria digitata* showed consistently lower similarity (0.09–0.27) across all chemotypes (Figure 9), collectively identifying *Undaria pinnatifida* as the species with the greatest overall structural similarity to known PPAR γ agonists.



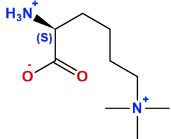
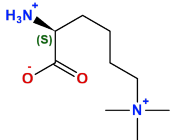
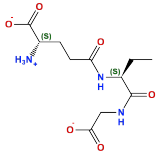
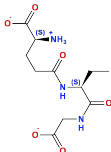
The similarity between *Undaria pinnatifida* and fatty acid/lipid-like agonists was driven exclusively by fatty acid metabolites, while their similarity to natural product-derived agonists was primarily driven by terpenoids, with additional contributions from steroids, phenols, fatty acids and nitrogen-containing compounds; the similarity between *Saccharina latissima* and natural product-derived agonists was attributable mainly to carotenoids, with minor contributions from phenols and carbohydrates (Figure 10).



3.4.2 Consensus pharmacophore screening identifies candidate PPAR γ agonists across multiple chemotypes

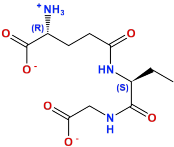
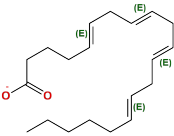
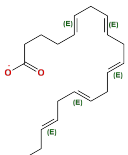
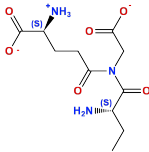
Pharmacophore screening identified 11 metabolites satisfying the RMSD < 2.0 Å acceptance threshold, of which 9/11 originated from *Undaria pinnatifida* and 2/11 from *Laminaria digitata*, with no hits identified for *Saccharina latissima* (Table 2). The strongest pharmacophore matches were observed for two nitrogen-containing *Undaria pinnatifida* metabolites, which achieved RMSD values of 0.364 Å and 0.879 Å.

Table 2: Seaweed metabolites satisfying the consensus PPAR γ pharmacophore screening criteria (RMSD < 2.0). Metabolites are ordered according to ascending pharmacophore RMSD.

2D structure	LOTUS ID	Species	Chemotype	RMSD (Å)
	LTS0214051	<i>Undaria pinnatifida</i>	Nitrogen-containing compounds	0.364
	LTS0143093	<i>Undaria pinnatifida</i>	Nitrogen-containing compounds	0.879
	LTS0108303	<i>Undaria pinnatifida</i>	Steroids	1.150
	LTS0187229	<i>Undaria pinnatifida</i>	Fatty acids	1.166

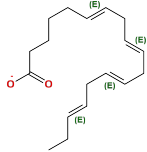
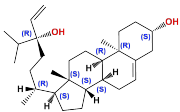
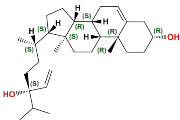
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Table 2 – continued from previous page

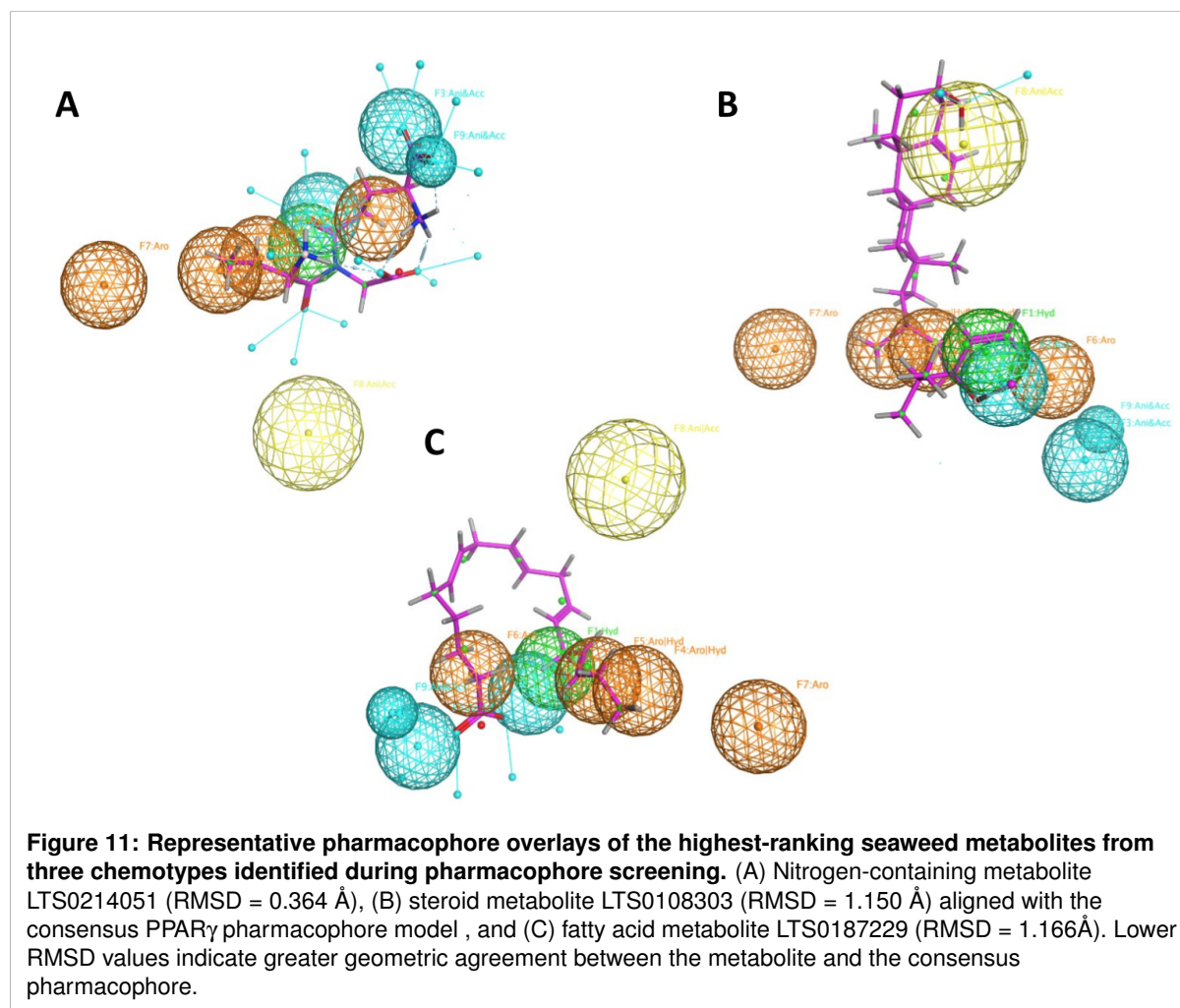
2D structure	LOTUS ID	Species	Chemotype	RMSD (Å)
	LTS0259881	<i>Undaria pinnatifida</i>	Fatty acids	1.169
	LTS0029907	<i>Undaria pinnatifida</i>	Nitrogen-containing compounds	1.184
	LTS0202199	<i>Undaria pinnatifida</i>	Nitrogen-containing compounds	1.185
	LTS0023647	<i>Undaria pinnatifida</i>	Steroids	1.303

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Table 2 – continued from previous page

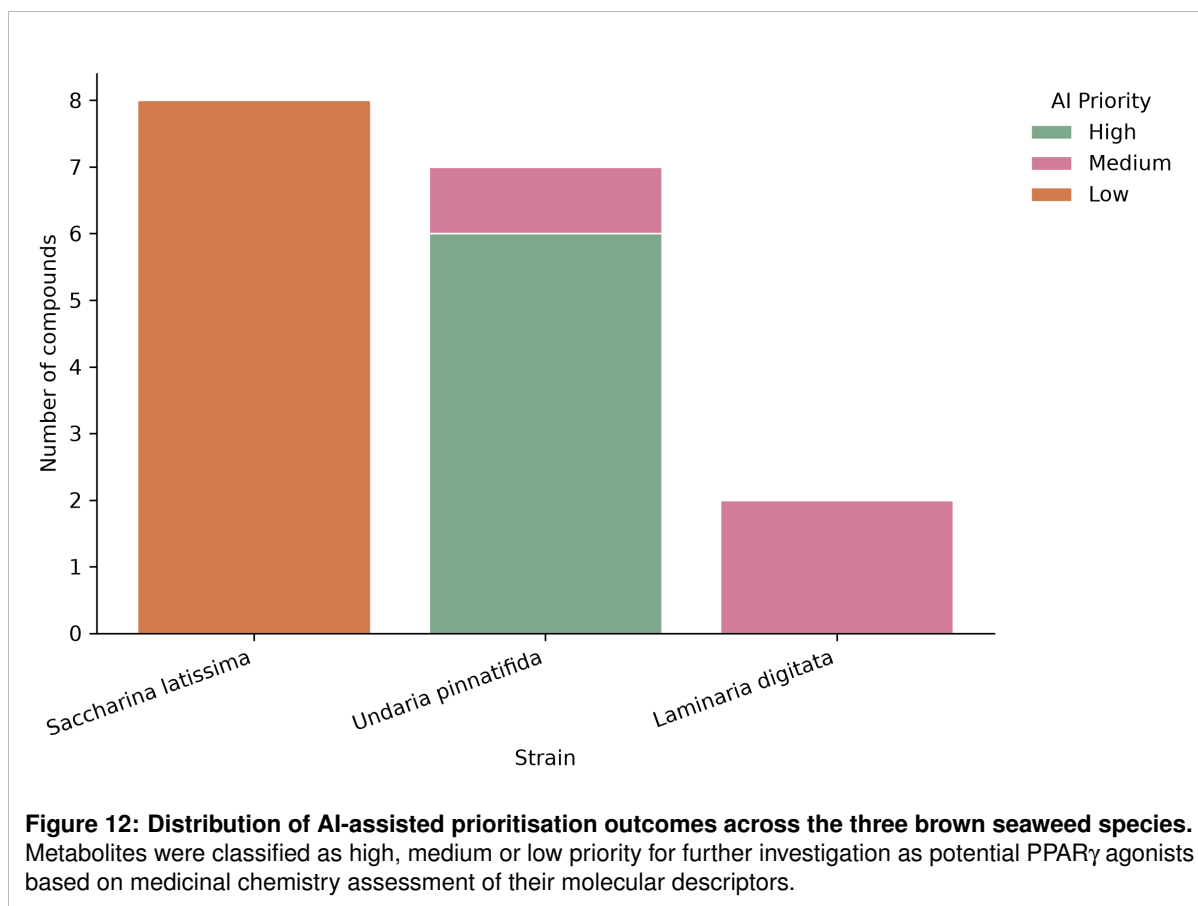
2D structure	LOTUS ID	Species	Chemotype	RMSD (Å)
	LTS0271804	<i>Undaria pinnatifida</i>	Fatty acids	1.349
	LTS0033475	<i>Laminaria digitata</i>	Nitrogen-containing compounds	1.506
	LTS0174680	<i>Laminaria digitata</i>	Nitrogen-containing compounds	1.506

Representative pharmacophore overlays revealed that seaweed metabolites from multiple chemotypes, including nitrogen-containing compounds, fatty acids and steroids, aligned with the consensus pharmacophore despite their structural diversity (Figure 11).



3.4.3 AI-assisted prioritisation identifies *Undaria pinnatifida* as the most promising source of candidate agonists

AI-assisted prioritisation classified all eight *Saccharina latissima* metabolites as low priority, six of the seven prioritised *Undaria pinnatifida* metabolites as high priority (one medium), and both *Laminaria digitata* metabolites as medium priority, with no high-priority candidates identified for either of the latter two species (Figure 12)



3.5 Molecular Docking Independently Validates Prioritised Compounds

To provide an independent, structure-based check on these ligand-based rankings, prioritised metabolites were evaluated by molecular docking against the PPAR γ ligand-binding pocket.

3.5.1 Re-docking validates the molecular docking protocol

Re-docking of the co-crystallised reference ligand achieved a docking score of -8.47 kcal/mol and a pose RMSD of 2.02 Å, reproducing three of the nine conserved experimental interactions (33%) and validating the docking protocol for subsequent screening (Figure 13).

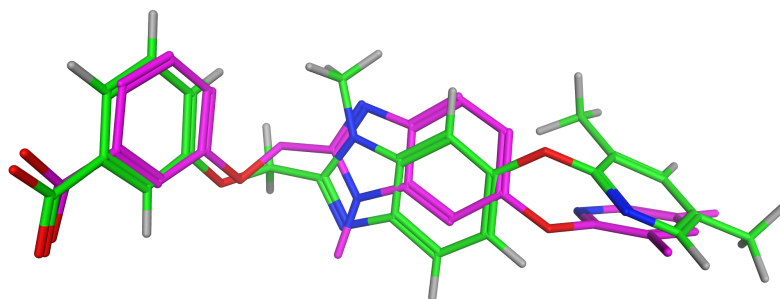
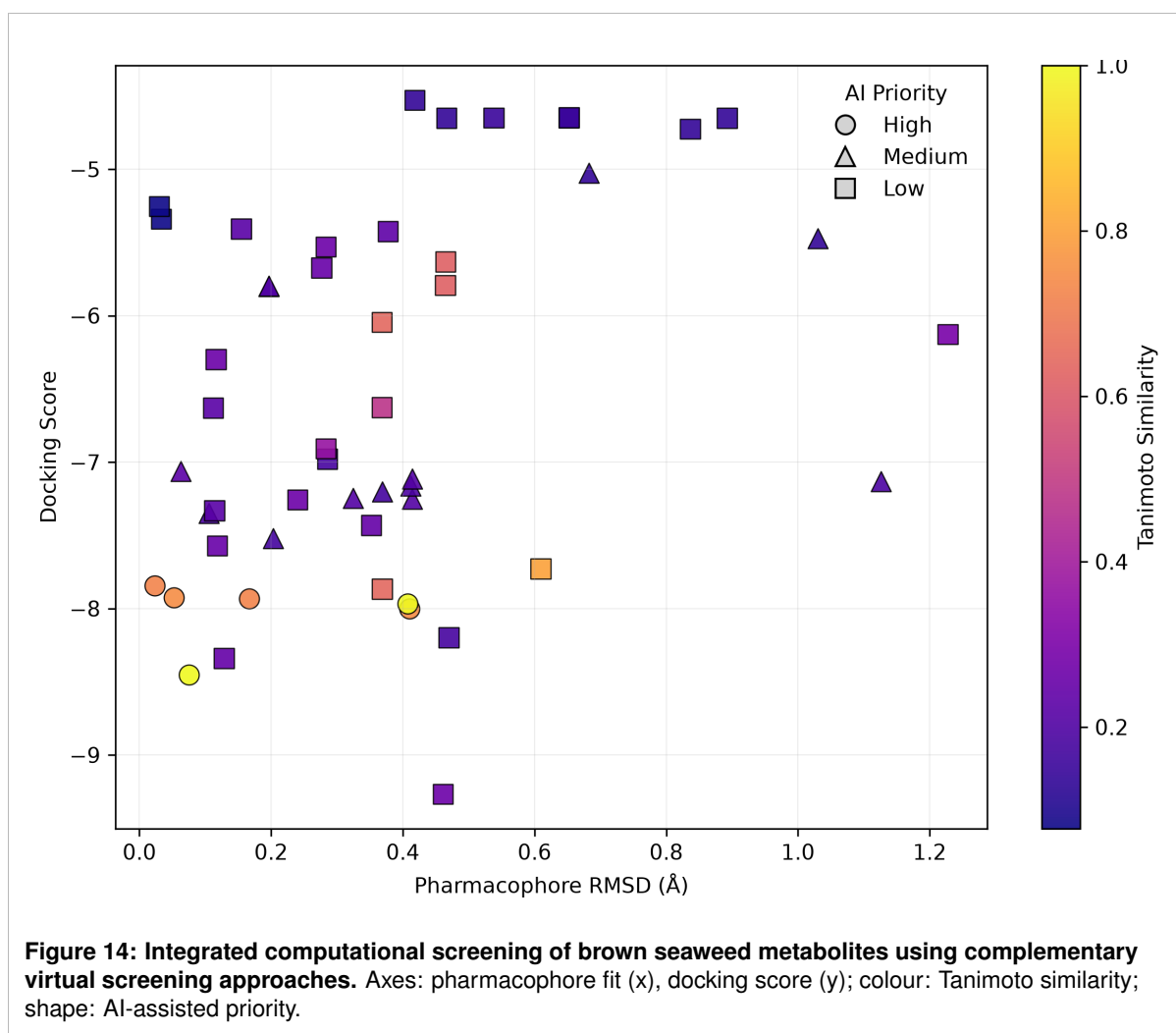


Figure 13: Structural overlay of the crystallographic and re-docked PPAR γ reference ligands. The experimentally determined ligand (magenta) and the selected re-docked pose (green) are superimposed within the PPAR γ ligand-binding pocket to demonstrate agreement between the experimental and predicted binding modes.

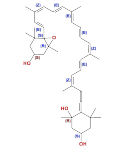
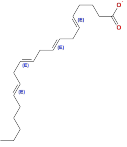
3.5.2 Integration of Computational Screening Methods Reveals Consistent Candidate Prioritisation

Integrated visualisation of the computational screening results demonstrated that the highest-ranking docking candidates were generally supported by more than one prioritisation strategy rather than a single screening method (Figure 14). High AI-priority compounds (circles) were predominantly associated with more favourable docking scores, while compounds exhibiting the greatest structural similarity to known PPAR γ agonists (yellow/orange markers) also clustered among the strongest docking candidates. Conversely, several compounds with favourable pharmacophore alignment (low RMSD) produced only moderate docking scores, and a number of well-docked compounds exhibited relatively low Tanimoto similarity, indicating that structural similarity, pharmacophore matching and molecular docking capture distinct aspects of ligand recognition. A Pearson correlation analysis identified a weak positive correlation between pharmacophore RMSD and docking score ($r = 0.313$, $p = 0.001$), indicating a statistically significant but modest relationship between the two screening metrics. Collectively, these findings demonstrate that integrating multiple computational screening approaches provides greater confidence for candidate prioritisation than reliance on any individual method alone.



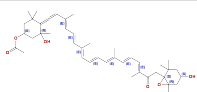
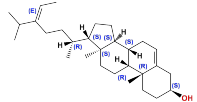
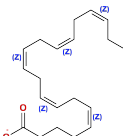
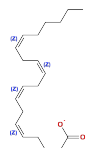
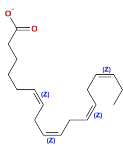
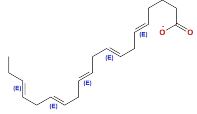
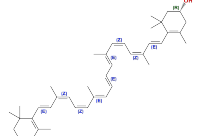
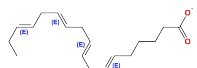
To identify the specific metabolites driving this enrichment, the ten highest-ranking compounds by docking score were examined across all screening outputs. Their pharmacophore RMSD values, maximum Tanimoto similarities and AI-assisted priority classifications are summarised in Table 3.

Table 3: Top 10 candidate seaweed metabolites ranked according to molecular docking score against PPAR γ . Corresponding Tanimoto similarity, pharmacophore RMSD and AI prioritisation are shown for comparison across the integrated virtual screening workflow.

2D structure	LOTUS ID	Species	Scaffold	Tanimoto	RMSD (Å)	Docking score	AI
	LTS0227522	<i>Saccharina latissima</i>	Carotenoids	0.268	0.461	-9.267	Low
	LTS0271804	<i>Undaria pinnatifida</i>	Fatty acids	1.000	0.076	-8.454	High

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Table 3 – continued from previous page

2D structure	LOTUS ID	Species	Scaffold	Tanimoto	RMSD (Å)	Docking score	AI
	LTS0149370	<i>Saccharina latissima</i>	Carotenoids	0.247	0.129	-8.340	Low
	LTS0183377	<i>Undaria pinnatifida</i>	Steroids	0.180	0.470	-8.199	Low
	LTS0174675	<i>Undaria pinnatifida</i>	Fatty acids	0.750	0.410	-8.003	High
	LTS0241153	<i>Undaria pinnatifida</i>	Fatty acids	1.000	0.408	-7.969	High
	LTS0228733	<i>Undaria pinnatifida</i>	Fatty acids	0.724	0.167	-7.934	High
	LTS0259881	<i>Undaria pinnatifida</i>	Fatty acids	0.750	0.053	-7.927	High
	LTS0261333	<i>Saccharina latissima</i>	Carotenoids	0.640	0.369	-7.867	Low
	LTS0187229	<i>Undaria pinnatifida</i>	Fatty acids	0.724	0.024	-7.846	High

3.5.3 Top-ranked candidate metabolites reproduce the crystallographic PPAR γ binding orientation

The five highest-ranked docked candidate metabolites were superimposed with the crystallographic PPAR γ agonist to compare their predicted binding orientations (Figure 15). All five candidates occupied the same ligand-binding region as the crystallographic ligand. The greatest structural overlap was observed within the hydrophobic ligand-binding cavity, while the fatty acid candidates additionally showed alignment of their terminal carboxylate groups with the corresponding region of the crystallographic ligand. Differences between the candidate metabolites were primarily observed in the orientation of the terminal side chains.

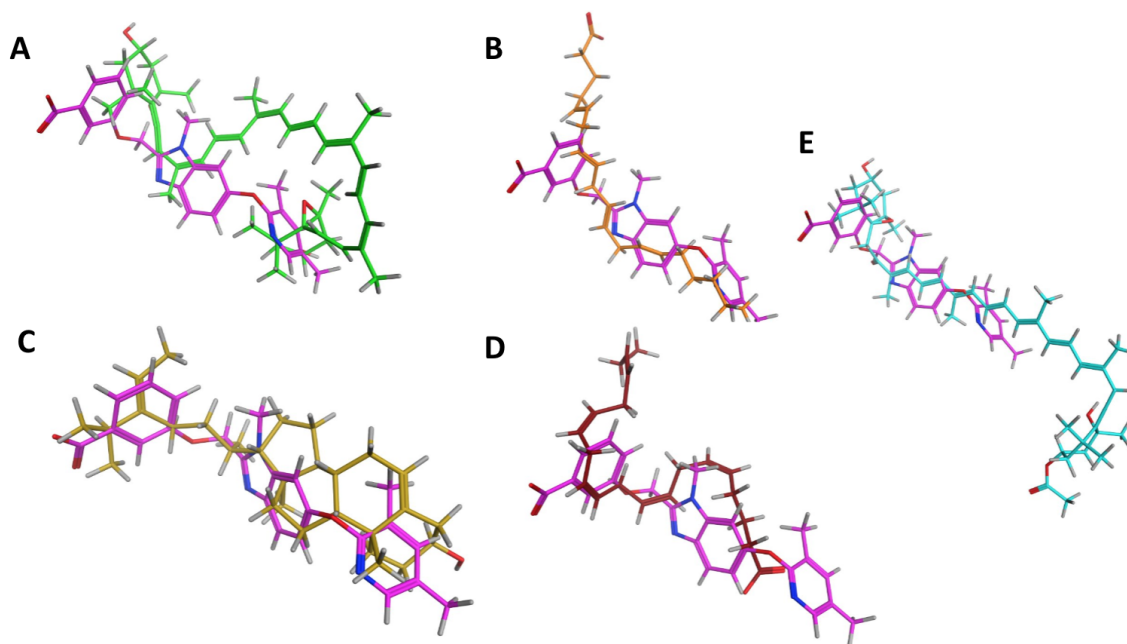


Figure 15: Structural comparison of the five highest-ranked docked seaweed metabolites with the crystallographic PPAR γ agonist. The crystallographic ligand is shown in magenta, while the docked candidate metabolites are shown as (A) LTS0227522 (carotenoid), (B) LTS0271804 (fatty acid), (C) LTS0183377 (steroid), (D) LTS0174675 (fatty acid) and (E) LTS0228733 (carotenoid).

4 Discussion

A representative reference database of experimentally validated PPAR γ agonists was established, encompassing the major chemotypes reported in the literature, including non-TZD synthetic compounds, TZDs, fatty acids and natural product-derived agonists [34]. The predominance of non-TZD synthetic ligands reflects two decades of medicinal chemistry optimisation to improve potency and safety over classical TZDs [35], whereas the inclusion of natural product-derived agonists demonstrates that structurally diverse natural compounds can satisfy the molecular requirements for PPAR γ activation [36]. This provided a biologically relevant framework for screening brown seaweed metabolites. Among the three seaweed species, *Undaria pinnatifida* displayed the greatest chemical diversity, whereas *Saccharina latissima* was largely dominated by carotenoids and *Laminaria digitata* contained comparatively few reported metabolites.

Surface mapping confirmed the characteristic hydrophobic, Y-shaped ligand-binding pocket of PPAR γ [12], while ligand interaction analysis identified a conserved interaction network involving CYS285, SER289, ARG288, HIS323, MET364 and HIS449 across structurally distinct chemotypes. This suggests that receptor recognition is driven primarily by conserved molecular interactions rather than overall scaffold similarity [37]. Although TYR473 interacted less frequently than expected despite its established role in stabilising helix 12 [38], this is consistent with growing interest in selective/partial PPAR γ modulators that avoid full helix-12 stabilisation while retaining activity and reducing adverse effects [12][39]. Although highly active agonists were predominantly synthetic compounds, this likely reflects the historical emphasis of medicinal chemistry on synthetic ligand optimisation [40] rather than an inherent superiority of synthetic scaffolds. In contrast, fatty acids – recognised endogenous PPAR γ ligands [41] – emerged as the strongest candidates during computational screening despite being underrepresented among the highest-potency reference agonists. Together, the binding pocket analysis, ligand interaction mapping and SAR analysis informed construction of a consensus nine-feature pharmacophore that closely resembled previously reported PPAR γ pharmacophore models [42].

The integrated screening workflow combined structural similarity, pharmacophore screening and AI-assisted prioritisation, allowing compounds supported by multiple independent approaches to be prioritised with greater confidence than any single method alone [43]. Fatty acids from *Undaria pinnatifida* were the only chemotype consistently prioritised by all three approaches, a finding with strong biological plausibility given that fatty acids are endogenous PPAR γ ligands [44] and that *Undaria pinnatifida* is recognised as a rich source of polyunsaturated fatty acids such as EPA [45][46]. This observation was strengthened by the identification of 5,8,11,14-eicosatetraenoic acid, which exhibited an exact structural match (Tanimoto similarity = 1.00) to the reference agonist arachidonic acid (ACD). As ACD and related polyunsaturated fatty acids act as endogenous PPAR γ ligands and serve as precursors of the potent

endogenous agonist 15-deoxy- $\Delta^{12,14}$ -prostaglandin J₂ (15d-PGJ₂), this finding provides additional biological support for prioritising seaweed-derived fatty acids [41].

Pharmacophore screening further suggested that chemically distinct scaffolds, including nitrogen-containing compounds, fatty acids and steroids, could satisfy the same conserved pharmacophoric requirements through different structural architectures, consistent with the large and permissive PPAR γ ligand-binding pocket [12]. Despite having no knowledge of the experimentally derived pharmacophore, AI-assisted prioritisation independently converged on the same lipid-like structural characteristics, supporting its potential as a complementary early-stage prioritisation tool when used alongside conventional cheminformatics approaches [47]. However, AI prioritised only the fatty acid chemotype and did not identify the broader range of candidate scaffolds highlighted by pharmacophore screening or molecular docking, suggesting that descriptor-based prioritisation alone may be biased towards compounds with physicochemical properties resembling established endogenous ligands. Similarly, Tanimoto similarity alone proved insufficient, as carotenoids displayed high structural similarity to natural product agonists but failed to satisfy the complete pharmacophore [48]. Collectively, these findings underscore the value of combining structural similarity, pharmacophore matching, AI-assisted prioritisation and molecular docking to capture this diversity. Nevertheless, the apparent enrichment of fatty acids within *Undaria pinnatifida* should be interpreted cautiously, as lipid-rich metabolites from *Saccharina latissima* remain comparatively under-represented within the curated database [49].

The molecular docking protocol successfully reproduced the experimental binding mode of the co-crystallised ligand [50][51], supporting the suitability of the docking protocol for virtual screening. The weak correlation between pharmacophore RMSD and docking score [52] further suggested that the two methods evaluate different aspects of ligand recognition, supporting their complementary application within the workflow. Importantly, all three consensus compounds identified by Tanimoto similarity, pharmacophore screening and AI-assisted prioritisation ranked among the ten highest-scoring docked metabolites, providing independent computational support for their prioritisation while not constituting evidence of agonist activity [53]. These compounds consistently conserved interactions with ARG288, supporting the importance of the negative ionisable pharmacophore feature [54], together with partial conservation of SER289 and CYS285 [55]. Collectively, these findings suggest that a lipid-like scaffold incorporating an extended hydrophobic region and terminal polar anchor represents a favourable structural framework for PPAR γ recognition.

Docking additionally identified promising binders that were not prioritised by the preceding screening methods, including two *Saccharina latissima* carotenoids and one *Undaria pinnatifida* steroid, demonstrating that docking contributed complementary information rather than simply reinforcing previous predictions. Among these, the carotenoid neoxanthin (LTS0227522) achieved the most favourable docking score of all screened metabolites, exceeding that of the co-crystallised reference ligand. However, dock-

ing scores should not be interpreted as direct measures of binding affinity or agonist potency, as current scoring functions are known to favour larger, more lipophilic molecules because additional molecular contacts generally improve calculated scores [19][56]. Consequently, neoxanthin's high ranking may partly reflect its large, highly conjugated scaffold rather than superior biological activity. Nevertheless, its favourable binding pose highlights neoxanthin as a promising candidate for further investigation using molecular dynamics simulations and experimental validation. Given evidence that PPAR γ activation upregulates FXR signalling [14], these candidates may hold therapeutic relevance beyond direct PPAR γ activity, providing further rationale for their experimental validation.

Several limitations should be considered when interpreting these findings. Firstly, approximately 60% of the curated seaweed metabolites originated from *Undaria pinnatifida*, while the reference agonist database was dominated by synthetic ligands, introducing potential bias during compound prioritisation. Expanding both datasets with additional marine metabolites and experimentally validated natural-product agonists would improve the robustness of future analyses. Secondly, chemically distinct scaffold classes were compared using the same virtual screening metrics, despite differences in molecular size, flexibility, and binding behaviour. Since many docking scoring functions can favour larger molecules [19][56], this may introduce bias when comparing different chemotypes. Therefore, analysing each scaffold class separately may provide a more balanced evaluation. Scaffold-specific analyses may therefore provide a more balanced comparison between chemotypes. Finally, all computational methods employed here represent static protein–ligand interactions and cannot fully capture receptor flexibility or ligand dynamics [57]. Likewise, the consensus pharmacophore was derived from a single binding hypothesis despite evidence that PPAR γ can accommodate multiple ligand-binding conformations [58]. Future work should therefore incorporate molecular dynamics simulations, ensemble pharmacophore approaches and experimental biochemical and cell-based transactivation assays to validate receptor activation and therapeutic potential [53].

5 Conclusion

This study developed an integrated computational workflow combining database curation, binding pocket characterisation, structure–activity relationship analysis, pharmacophore modelling, Tanimoto similarity, AI-assisted prioritisation and molecular docking to identify candidate PPAR γ modulators from brown seaweed metabolites.

Fatty acids consistently emerged as the highest-priority chemotype across all screening methods, while pharmacophore screening revealed that structurally diverse scaffolds could satisfy the conserved molecular features required for PPAR γ recognition. Molecular docking independently supported the prioritised compounds while identifying additional candidate chemotypes.

Overall, the workflow provides a robust framework for prioritising marine-derived PPAR γ modulators and supports the potential of seaweed metabolites as a source of novel anti-inflammatory leads. Future work should incorporate molecular dynamics simulations and experimental validation to confirm receptor activation and therapeutic potential.

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